MIL-C-13609B(PA)
25 November 1974
SUPERSEDING
MIL-C-13609A(MU)
14 July 1971

MILITARY SPECIFICATION

CASTING SOLVENTS FOR OIO, OGK AND AHH PROPELLANTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers three types of casting solvents for use in the manufacture of propellants.

1.2 Classification. The casting solvent shall be of the following types as specified:

Type I - Casting Solvent, OGK
Type II - Casting Solvent, OIO
Type III - Casting Solvent, AHH

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-N-246	- Nitroglycerin
MIL-T-301	- Triacetin (Glyceral Triacetate)
MIL-N-3399	- 2-Nitrodiphenylamine
MIL-A-48078	- Ammunition, Standard Quality Assurance Provision, General Specification for.

STANDARDS

MILITARY

MIL-STD-286	- Propellants, Solid: Sampling Examination and Testing

FSC: 1376

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Publication.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 Transportation, Parts 0-170 (see Tariff Number 27).

(The Interstate Commerce Commission Rules are now a part of the code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite 49 CFR 0-170 (latest revisions).)

3. REQUIREMENTS

3.1 Materials.-The raw materials used in the manufacture of casting solvents purchased under this specification shall conform with the requirements of the applicable specifications MIL-N-246, MIL-T-301, and MIL-N-3399.

3.2 Moisture.-The moisture present in Type I, Type II, and Type III casting solvent shall be 0.5 percent maximum, when tested as specified in 4.5.1.

3.3 Composition.-The composition of the casting solvents, on a moisture-free basis, shall be as given in Table I when tested as specified in 4.5.2.

Table I

	Type I	Type II	Type III
Nitroglycerin, percent (Type I)	72.0 ± 1.0	70.0 ± 1.0	75.0 ± 1.0
2-Nitrodiphenylamine, percent	1.0 ± 0.1	1.0 ± 0.1	1.0 ± 0.1
Triacetin, percent	27.0 ± 1.0	29.0 ± 1.0	24.0 ± 1.0

3.4 Stability.-The casting solvent shall not change the color of standard potassium iodide-starch paper in less than 10 minutes, when subjected to the 82.2°C heat test when tested as specified in 4.5.3.

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3.5 Workmanship.-Details of the manufacturing process and equipment used by the contractor shall be submitted in writing to the contracting agency and be approved in writing before the contractor proceeds with the manufacture of the casting solvent. Any deviation from the approved manufacturing process shall be submitted in writing to the contracting agency, and written approval received prior to use.

3.6 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract, (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 2 ounces of casting solvent obtained by sampling as described in 4.4.2. The samples shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All material shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be Performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.5 of this specification.

4.3.3 Rejection.-See MIL-A-48078.

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. For the material covered by this specification, a lot shall consist of one or more batches of casting solvent produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of casting solvent that has been subjected to the same chemical or physical process.

4.4.2 Sampling.-Two independent, representative samples of approximately 2 ounces each shall be removed from each batch. The casting solvent shall be transferred to a rubber bottle with a rubber stopper. The bottle shall be labeled to the show the following:

(a) Source of manufacture.
(b) Plant.
(c) Contract or purchase order number.
(d) Number of pounds in the lot.
(e) Lot number.

If either sample fails to comply with any of the requirements specified the batch shall be rejected. Caution should be exercised in handling the material due to its explosive hazards and toxicity (see 6.4).

4.4.2.1 Classification of Defects.-See MIL-A-48078.

Test	Test Classification
Moisture	Major B
Composition:	
Nitroglycerin	Major B
2-Nitrodiphenylamine	Major B
Triacetin	Major B
Stability	Major B
Workmanship	Major B

4.4.3 Inspection Equipment.-The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.4 and 4.5. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purpose.

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4.5 Test Methods and Procedures.-The tests in 4.5.1 through 4.5.5 shall be performed using prescribed analytical procedures for duplicate determinations given in standard analytical textbooks. (see 6.3).

4.5.1 Determination of Moisture Content.-The moisture content shall be conducted in accordance with Method 101.5 of MIL-STD-286 except that a visual end point shall be used.

4.5.2 Determination of Composition.

4.5.2.1 Nitroglycerin.-The nitroglycerin shall be determined in accordance with Method 208.1 of MIL-STD-286 except that a three gram sample shall be used.

4.5.2.2 Nitroglycerin (alternate method).-The nitroglycerin content may be determined in accordance with Method 510.2 of MIL-STD-286 except a complete detail description of the method shall be forwarded to the Commanding Officer, Attn: SARPA-QA-A-P, Picatinny Arsenal, Dover, NJ 07801 for approval.

4.5.2.3 2-Nitrodiphenylamine.-The 2-nitrodiphenylamine content shall be determined in accordance with Method 218.1 of MIL-STD-286.

4.5.2.4 Triacetin.-The percentage of triacetin in the sample shall be calculated on a moisture-free basis by subtracting from 100 percent the sum of the percentages of nitroglycerin and 2-nitrodiphenylamine.

4.5.3 Stability.-The stability shall be determined as follows: Filter a portion of the casting solvent sample through two thicknesses "S. and S." Number 604 filter paper or equivalent. Transfer a 2 ml portion of the filtered material, by means of a pipette, to each of three test tubes which are 5.5 inches long, 0.50 inches internal diameter and 0.62 inches outside diameter. (Care should be taken during the transfer not to leave droplets of the casting solvent on the sides of the test tubes.) Stopper each tube by means of a new, tightly fitting cork through which passes a tightly fitted glass rod equipped with a platinum holder for a strip of standard potassium iodide starch-indicator paper. Using forceps hang on to the platinum holder a strip of standard potassium iodide starch-indicator paper. The standard potassium iodide starch-indicator test paper shall normally be approximately 1 inch long and 3/8 inch wide. Moisten a horizontal section in the upper half of the standard test paper with a 50 percent solution (by volume) of pure glycerin in distilled water. This is conveniently accomplished by dip-

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ping a small diameter glass rod into the glycerin-water solution, and as the rod is withdrawn, contact is made with the side of the container so as to minimize the volume of the solution adhering to the rod. Draw the rod across the paper strip so as to produce a level and distinct line of demarcation on the lower edge of the wet area. Prepare a blank by suspending a moistened strip of the standard test paper in a clean dry tube. Adjust the temperature of the heat tube bath to 82.2 + 1°C. and insert all four test tubes into the heating solution. The depth of immersion of the test tubes into the heating solution should be approximately 2 inches. The bath should be placed in such a position that the test tubes are viewed against a white background illuminated by bright diffused daylight. Note the time of insertion of the tubes into the bath. During the test the line of demarcation on the standard test paper should be 3 inches above the level of the casting solvent in the test tube. The line of demarcation in the blank tube is regulated at an equivalent height in the tube. Consider the end point of the test to be the first appearance of any discoloration at the line of demarcation between the wet and the dry portion of the test paper in the sample tube which is in excess of the discoloration observed at the same position on the test paper in the blank. Note the time for completion of the test to the nearest minute. Consider the minimum time for the three tubes to represent the heat test value of the sample. After the test rinse all rods and tubes with acetone to remove casting solvent, wash with warm soapy water, rinse thoroughly with tap water, then with distilled water and dry in a steam oven at approximately 80°C.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C.-Casting solvent is classed as a "Forbidden Explosive" in Tariff Number 27, (Graziano Tariff), Code of Federal Regulations, Interstate Commerce Commission Regulations for the transportation of Explosives and Other Articles by Freight. For shipment by public highways see Tariff Number 27, appropriate section of Interstate Commerce Commission Regulations applying to shipments made by Way of Common and Contract Carriers by Public Highway.

5.2 Marking.-The marking of interior packages and exterior shipping containers shall be in accordance with instructions contained in Tariff Number 27, Interstate Commerce Commission Regulations, and in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended Use.-The OGK, OIO and AHH casting solvents are intended use in the manufacture of rocket grain propellant.

6.2 Ordering data.-a. MIL-A-48078.
b. Packing (see 5.1)
c. First Article (see 4.3)

6.3 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ATTN: SARPA-QA-A-P, Picatinny Arsenal, Dover, New Jersey, 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.4 Disposal of sample bottle.-The bottle shall be cleaned as follows:

(a) Rinse thoroughly with acetone.
(b) Flush generously with hot tap water (3 minutes under a direct flow is considered adequate).
(c) Rinse with distilled water and dry.

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